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Submitted to: Nebraska Department of Transportation (NDOT)

Project Goals:

The goal of the project is to convert an Excel-based workflow into a Python-based system. We need to replicate the mix design program and create a Python code that allows users to design parameters in a structured, step-by-step manner. Then the final design will be implemented as a weight chart for one cubic yard of concrete. NDOT gave us four requests: a) translate the Excel logic into Python, b) implement sequential user inputs, c) generate a final mix design output, and d) prepare and evaluate four concrete mix scenarios.

Project Tasks:

Task 1: Translate the Excel logic into Python

The Excel sheet uses a list of equations to find the volume of each mix ingredient for a cubic yard of concrete. In Python, we will make each equation as a function and add the given values in the excel.

Task 2: Implement sequential user inputs

A function will be made that asks the user for input. It takes it step by step, starting with the project and concrete class. Next, it will ask about the cementitious materials. Then it will find the water-cement ratio and air content. The specific gravity for each input is next. Finally, it will ask about the aggregate input.

Task 3: Generate a final mix design output

The next function will take the inputs of the user and calculate the volumes needed of each ingredient. It is combined to find the final batch volume.

Task 4: Prepare and evaluate four concrete mix scenarios

After research, we will find four concrete mix scenarios to test on our Python program. We will use different concrete mixes that can be input for our Python model and find the volumes needed for the ingredients and the final batch volume. It will be used to validate the workflow and learn more about different concrete mixes and the differences between different concrete mixes and the uses of different mixes.

Project deliverables:

The deliverables and due dates for this project are summarized in the following table:

Deliverable	**Due to Client by**
Github Link	2-24-26
Project Management	2-3-26
Technical Implementation	2-24-26
Written Technical Report	2-24-26